

Convolutional neural network framework for deepfake detection: A diffusion-based approach

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ABSTRACT

In the rapidly advancing domain of synthetic media, DeepFakes emerged as a potent tool for misinformation and manipulation. Nevertheless, the engineering challenge lies in detecting such content to ensure information integrity. Recent artificial intelligence contributions in deepfake detection have mainly concentrated around sophisticated convolutional neural network models, which derive insights from facial biometrics, including multi-attentional and multi-view mechanisms, pairwise/siamese, distillation learning technique and facial-geometry approaches. In this work, we consider a new diffusion-based neural network approach, rather than directly analyzing deepfake images for inconsistencies. Motivated by the considerable property of diffusion procedure of unveiling anomalies, we employ diffusion of the inherent structure of deepfake images, seeking for patterns throughout this process. Specifically, the proposed diffusion network, iteratively adds noise to the input image until it almost becomes pure noise. Subsequently, a convolutional neural network extracts features from the final diffused state, as well as from all transient states of the diffusion process. The comprehensive experimental analysis demonstrates the efficacy and adaptability of the proposed model, validating its robustness against a wide range of deepfake detection models, being a promising artificial intelligence tool for DeepFake detection.

1. Introduction

DeepFake technology, which involves the creation of synthetic media such as images, videos, and audio, is rapidly evolving primarily through the use of deep learning models such as Generative Adversarial Networks (GAN) (Pan et al., 2019). These models can almost seamlessly substitute the facial characteristics of a person in the original content with someone else's likeness, making it exceptionally challenging even for human experts to discern their authenticity (Nightingale et al., 2017; Rossler et al., 2019). This rapid advancement has raised concerns due to the technology's potential misuse, especially when is used as a tool for misinformation, manipulation, and harassment; therefore, posing threats to human rights (Agarwal et al., 2019; Pintelas and Pintelas, 2022).

This potential threat has led to increased interest in the field of DeepFake detection (Qian et al., 2020; Huang et al., 2021; Liang et al., 2023; Hsu et al., 2020). However, detecting these manipulated pieces of media is not straightforward. It is a complex classification problem, which challenges both machines and humans (Li et al., 2020; Rossler et al., 2019). While significant efforts (Tolosana et al., 2022; Pintelas et al., 2023; Dolhansky et al., 2020; Zhao et al., 2021; Bonettini et al.,

2021; Heo et al., 2021; Liang et al., 2023) have been devoted to detecting DeepFakes using techniques such as attention mechanisms, transformer architectures and geometric feature extraction, the potential of diffusion processes in enhancing feature extraction for detection has not been thoroughly explored. Diffusion processes, grounded in physics and biology, describe how elements disperse through a medium (Weigel et al., 2011; Metzler et al., 2014) and have been effectively employed in tasks like image synthesis, denoising and anomaly detection (Prasath, 2017). Motivated by the hypothesis that DeepFakes, might exhibit unusually rapid diffusion states due to artificial blending of features and display odd diffusion patterns as they do not maintain natural image coherence (Pan et al., 2019), the core of the proposed approach lies on simulating how the inherent structure of a deepfake image diffuses, searching for patterns in that process.

In this work, we propose a new Diffusion-based Convolutional Network framework called “DiffConvNet”, which extracts features via a diffusion process for enhancing a CNN-based model to detect DeepFakes. Utilizing a diffusion network the input image (in fact the encoded representation of the first layers of a CNN) is iteratively diffused in order to extract its diffused representation states. Then, via a CNN

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model, we exploit the transient and the final states of diffusion, in order to extract features during this diffusion process and finally perform the prediction. Unlike prior approaches, which primarily rely on advanced architectures to directly analyze input images, the proposed method leverages the iterative diffusion process to reveal hidden patterns indicative of manipulation. By integrating a diffusion-based framework directly into a CNN pipeline, this study aims to uncover subtle anomalies, enhance feature extraction and improve the robustness of DeepFake detection models.

More specifically, the proposed approach incorporates a diffusion procedure through the convolutional layers in order to leverage diffusion patterns. In detail, the CNN backbone is initially trained with respect to the deepfake classification loss and then, is “cut” into two parts (refer also to Fig. 1), forming a Shallow and a Deep Network (SN and DN). Next, a Diffusion Network (DiffN) is attached between these two network parts forming a SN-DiffN-DN end-to-end network. The SN aims to encode the initial image into a more robust and suitable form for the diffusion process. Then, the DiffN diffuses the encoded representation aiming to reveal deepfake patterns. Finally, the DN aims to extract these patterns by exploiting the final and transient states of the diffusion procedure, in order to perform the final prediction. Additionally, this SN-DiffN-DN network is fine-tuned end-to-end with respect to the DeepFake classification loss in order to synergistically optimize each main part of this network. This implies that the SN is expected to learn better representations for the diffusion layers as regards the deepfake detection, the diffusion network is expected to be optimized in a way to perform more efficient diffusions with respect to the deepfake detection, while the DN will be optimized in order to extract in a more efficient way patterns and features via the diffusion states.

In our numerical experiments ResNet18 was utilized as a CNN backbone due to its high speed-accuracy trade-off performance (He et al., 2016; Canziani et al., 2017). Nevertheless, the proposed approach can be easily integrated into any CNN backbone network, with minor modifications. In the ablations studies we also incorporated our approach into Xception (Chollet, 2017), and EfficientNet (Tan and Le, 2019) backbones since these networks serve as the main utilized networks for DeepFake detection (Bonettini et al., 2021; Zhao et al., 2021).

The main contributions of this research work are summarized as follows:

- We propose the idea of leveraging the diffusion process to enhance the feature extraction capabilities of CNN models for detecting DeepFakes. Building on this idea, we present a new diffusion-based CNN framework, which efficiently incorporates the diffusion process into state-of-the-art CNN architectures, offering a flexible and adjustable framework for improved DeepFake detection.
- We propose a CNN-based model, which achieves superior performance, compared to state-of-the-art models, for the DeepFake detection problem.
- Finally, we conduct a comprehensive experimental analysis providing insights and interpretations about the reasons of the proposed model’s performance efficacy which strengthens the proposed method trustfulness.

The rest of this paper is followed by the recent related work for deepfake detection (Section 2), the detailed presentation of the proposed approach (Section 3), the case-study datasets used in our experiments (Section 4), followed by extensive comparison, ablation and visualization studies (Sections 5, 6 and 7 respectively). Finally, our discussions (Section 8) and conclusions (Section 9) offer insights into the effectiveness and potential future directions of proposed methodology in the field of DeepFake detection.

2. Related work

In this section, we summarize a wide range of state-of-the-art CNN-based deepfake detection methodologies. These methods are categorized into feature-based, attention-based, and training-based frameworks, all of which aim to improve detection by leveraging specific properties of the input data.

Feature-Based Frameworks: Feature-based techniques focus on extracting meaningful representations from images to distinguish real and manipulated content. Fine-Tuning (FT) methods (Huang et al., 2021) serve as a baseline, adapting pre-trained CNN backbones to the deepfake classification task. Facial-Fusion (Tolosana et al., 2022) further refine this process by analyzing segmented facial regions and employing fusion techniques across first and second-generation deepfakes, enhancing detection robustness. Facial geometry-based approaches (Liang et al., 2023) embed biometric features derived from facial landmarks into CNN-LSTM networks, leveraging spatial and frequency domain correlations for improved detection. The Multi-View approach (Pintelas et al., 2023) focuses on generating diverse perspectives of input images using mathematical transformations, enriching feature representations through the fusion of pre-trained CNN-extracted features. This technique has demonstrated improved performance in detecting subtle manipulation signs, making it effective for deepfake classification tasks.

Attention-Based Frameworks: Attention mechanisms have been pivotal in enhancing deepfake detection. Siamese-Attention techniques (Bonettini et al., 2021) integrate siamese networks with attention maps to refine feature spaces, optimizing them for fine-grained classification. Multi-attentional networks (Zhao et al., 2021) extend this concept by employing multiple spatial attention heads and textural enhancement blocks to capture subtle anomalies within manipulated regions, merging low-level textures and high-level semantics.

Training-Based Frameworks: Training-based frameworks investigate advanced training schemes to enhance the robustness and generalization of deepfake detection models. Pairwise learning strategies (Hsu et al., 2020) employ a two-step process using contrastive loss to train CNN models on pairs of images, learning a relational feature space. In the second step, a classifier based on this feature space is trained with binary cross-entropy loss to predict individual labels. Distillation-based models (Heo et al., 2021) utilize a teacher-student training scheme, where a pre-trained teacher model transfers knowledge to a smaller student model through distillation loss. This design integrates features from CNNs with a patch-based positioning model, which was then trained to interact with all positions in order to accurately identify artifact regions; thereby, addressing the false negative problem prevalent in previous methods.

In contrast to these approaches, which mainly emphasize on extracting features from the input images through complicated architectures, the proposed approach mainly differs in extracting features via the diffusion of input images. This approach is inspired by the hypothesis that manipulated images will exhibit unique patterns during the diffusion process, which can be critical for detecting deepfakes. By incorporating diffusion-based dynamics directly into the network, our method ensures compatibility across a variety of CNN backbones. Additionally, the diffusion patterns can offer insights into the detection process, enabling a deeper understanding of how deepfakes are identified. Detailed experimental analysis and extended ablation studies reveal the efficacy of the proposed approach.

3. Proposed framework

Next, we present the proposed approach starting from an abstract overview description, to end up into a comprehensive and detailed architectural description.

Fig. 1 illustrates a high-level description of the proposed DiffConvNet model. Initially, a classification neural-network model, using as backbone a pre-trained feature extraction network, is fine-tuned with

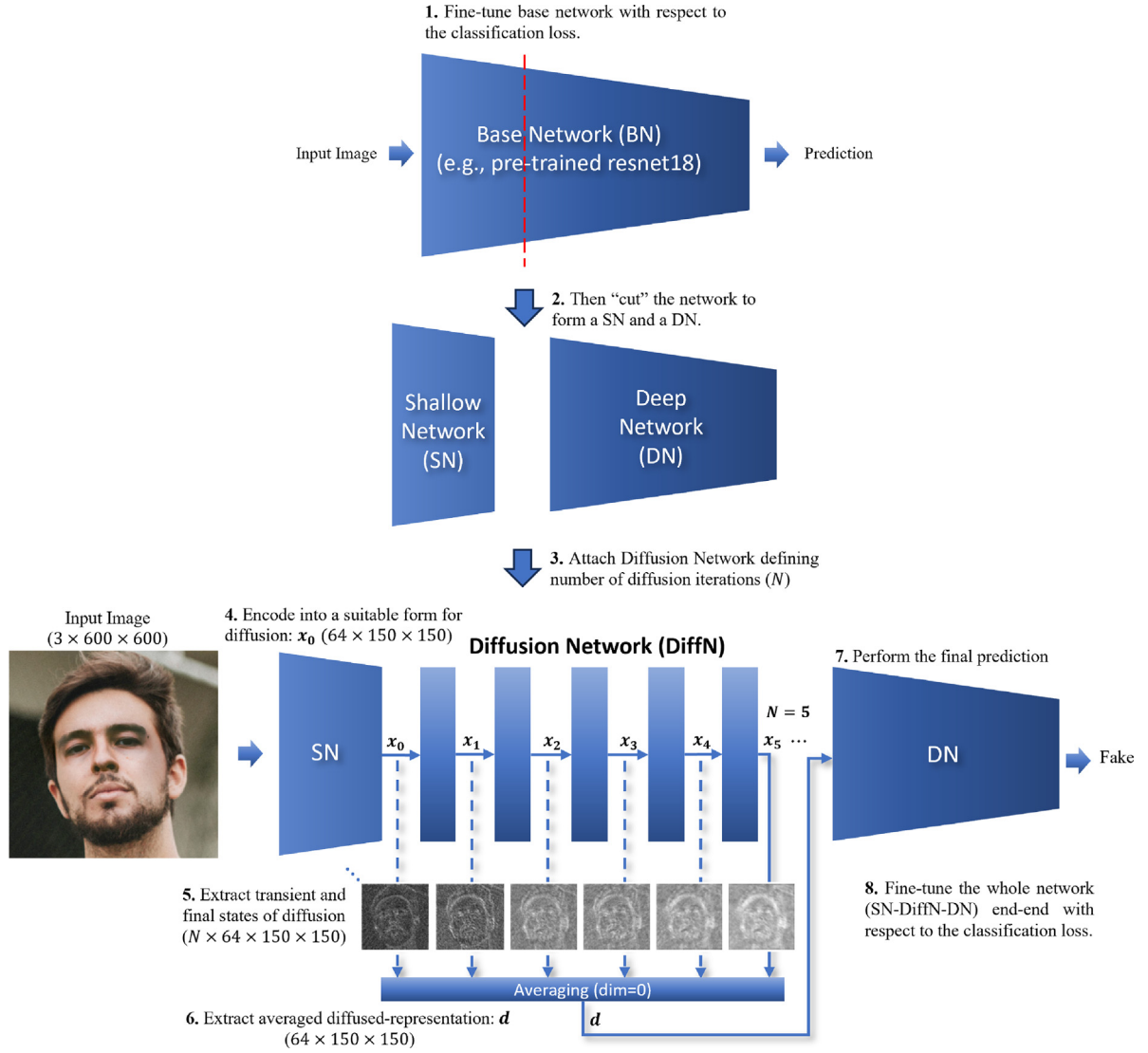


Fig. 1. High-level architectural overview of the DiffConvNet model. The proposed framework integrates a diffusion network between the shallow and deep parts of a pre-trained CNN backbone to enhance DeepFake detection by leveraging diffusion-based feature extraction.

respect to the classification loss. Notice that, this backbone network is composed by a series of modules of convolution layers followed by a classification head (series of dense layers along with the classification output layer). Then, this base network (BN) model is split into two network parts, forming a Shallow Network (SN) and a Deep Network (DN).

Next, the Diffusion Network (DiffN) constituted by convolutional layers is attached between the SN and DN. The SN has the objective to process the input image, converting it into an encoded and more robust representation (denoted as x_0) suitable for diffusion. Next, the encoded input x_0 is fed into the DiffN, which iteratively diffuses it for N iterations. Then, the x_0 , the transient diffused representations $x_1, x_2, \dots, x_{N-1}$ and the final diffusion state x_N are averaged in order to form a comprehensive diffused-representation d (it actually contains in a compressed form information of the diffusion process). Finally, the DN extracts patterns from the diffused representation d and performs the final prediction. In the sequel, we provide a detailed description of the proposed framework focusing on its main steps and its architectural design.

3.1. Description of the proposed framework

Initially, a pre-trained base network (BN) is fine-tuned with respect to the DeepFake Classification loss $\mathcal{L}$ defined as:

$$\mathcal{L}(\hat{y}, y; \Theta_{BN}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y}) \quad (1)$$

where y and $\hat{y}$ denotes the true and predicted label, respectively (Real or Fake), while Θ_{BN} represents the parameters of the base network. Thus, given an input image I , the fine-tuned function F of BN can be represented as:

$$F(I; \Theta_{BN}) \rightarrow \hat{y} \quad (2)$$

Subsequently, the fine-tuned BN is split to form the shallow network (SN) and the deep network (DN). Notice that Θ_{SN} and Θ_{DN} denote the parameters of the swallow and deep network, respectively, namely $\Theta_{BN} = [\Theta_{SN}, \Theta_{DN}]$. Furthermore, we employ the Diffusion Network (DiffN) for N iterations and attach it between the SN and DN; therefore, forming DiffConvNet model with parameters:

$$\Theta_{\text{DiffConvNet}} = [\Theta_{SN}, \Theta_{\text{DiffN}}, \Theta_{DN}] \quad (3)$$

Given an input image I , the initial encoded representation x_0 is extracted from the fine-tuned shallow network namely:

$$I \xrightarrow[\Theta_{\text{SN}}]{\text{SN}} x_0 \quad (4)$$

while through the DiffN with N iterations, the diffusion process is defined as:

$$x_0 \xrightarrow[\Theta_{\text{DiffN}}]{\text{DiffN}} x_N \quad (5)$$

where x_N is the output through N diffusion iterations and $x_1, x_2, \dots, x_{N-1}$ are the transient diffusion states in $1, 2, \dots, N-1$ iterations, respectively.

The diffusion process propagates through the network by iteratively applying the convolutional operations within the DiffN to the encoded representation x_0 generated by the SN. During each iteration, the representation evolves into transient states ($x_1, x_2, \dots, x_{N-1}$) by progressively diffusing its features. These states capture in a dynamic manner the patterns and anomalies, with the final state x_N being the outcome of the full diffusion process. The transient states, along with x_0 and x_N , are aggregated via averaging to create the diffused representation d (Eq. (6)), which is then fed into the DN for final classification (Eq. (7)).

$$d = \frac{1}{N} \sum_{i=0}^N x_i \quad (6)$$

$$d \xrightarrow[\Theta_{\text{DN}}]{\text{DN}} \hat{y} \quad (7)$$

Finally, based on the same classification loss $\mathcal{L}$ defined in Eq. (1), we fine-tune the DiffConvNet model. Therefore, given the input image I , the fine-tuned function of the network DiffConvNet can be represented as:

$$F(I; \Theta_{\text{SN}}, \Theta_{\text{DiffN}}, \Theta_{\text{DN}}) \rightarrow \hat{y} \quad (8)$$

Algorithm 1 summarizes the proposed approach in a compact form.

The fine-tuning process of the DiffConvNet model involves optimizing the parameters of the SN, DiffN and DN jointly during the end-to-end training. The DeepFake classification loss $\mathcal{L}$ is computed based on the final prediction of the DN. This loss is back-propagated through the entire network, allowing each component to adjust its parameters in a synergistic manner. Specifically, the SN learns to generate encoded representations x_0 that are more suitable for the diffusion process, while the DiffN parameters are optimized to reveal subtle patterns and anomalies through the iterative diffusion process. The DN, in turn, adapts its parameters to effectively extract features from the diffused representation d and improve classification performance. By fine-tuning the entire network end-to-end, the SN, DiffN and DN components are collaboratively optimized to enhance the detection of DeepFakes, ensuring that the entire architecture works in harmony for this task.

3.2. Architecture

Finally, for completeness Tables 1 and 2 present the architecture of BN and DiffConvNet, respectively, using ResNet18 as backbone. This architectural description serves to provide a comprehensive overview of the network components and their configurations, setting a clear understanding of the systemic implementations made in this study.

The modules from Layer 0 to Layer 4 correspond to the structured design and pre-trained weights inherent to the ResNet18 backbone, while global average pooling (GAP) and fully connected (FC) head are used in order to adapt to the task of deepfake classification. Each of the designated Layers is constructed utilizing the fundamental block design of the backbone, constituted with operations including convolutional layers, batch normalization and ReLU activation functions. Notice that H and W denote the height and width dimensions of the input RGB image, respectively. For the purpose of this presentation, we have focused primarily on showcasing the dimension reduction resulting

Algorithm 1 Pseudocode description of proposed framework.

Inputs:

BN: A Backbone network constituted by modules of convolution layers followed by a classification head.

DiffN: A convolution layer repeated N times (as the number of diffusion iterations).

Output:

DiffConvNet: Proposed DiffConvNet classification model.

- 1: Fine-tune BN with respect to DeepFake classification loss $\mathcal{L}$
- 2: Split the BN to form the SN and DN ▷ Formulation of DiffConvNet
- 3: Attach DiffN into the SN and DN to form the DiffConvNet
- 4: Use SN to encode I into x_0 ▷ DiffConvNet inner prediction mechanism
- 5: Use DiffN to diffuse x_0 for N iterations and extract $x_1, \dots, x_N$
- 6: Average $x_0, \dots, x_N$ to create d
- 7: Use DN to encode d and predict $\hat{y}$
- 8: Fine-tune DiffConvNet with respect to DeepFake classification loss $\mathcal{L}$

Table 1

Architectural presentation of BN for ResNet18 backbone.

	modules	input	output
Base Network (BN)	Layer 0	$3 \times H \times W$	$64 \times \frac{H}{4} \times \frac{W}{4}$
	Layer 1	$64 \times \frac{H}{4} \times \frac{W}{4}$	$64 \times \frac{H}{4} \times \frac{W}{4}$
	Layer 2	$64 \times \frac{H}{4} \times \frac{W}{4}$	$128 \times \frac{H}{8} \times \frac{W}{8}$
	Layer 3	$128 \times \frac{H}{8} \times \frac{W}{8}$	$256 \times \frac{H}{16} \times \frac{W}{16}$
	Layer 4	$256 \times \frac{H}{16} \times \frac{W}{16}$	$512 \times \frac{H}{32} \times \frac{W}{32}$
	GAP	$512 \times \frac{H}{32} \times \frac{W}{32}$	512
	FC Head	512	1

Table 2

Architectural presentation of DiffConvNet for ResNet18 backbone.

	modules	input	output
DiffConvNet	SN	Layer 0	$3 \times H \times W$
	DiffN	DiffLayer($\times N$)	$64 \times \frac{H}{4} \times \frac{W}{4}$
		Averaging (dim=0)	$N \times 64 \times \frac{H}{4} \times \frac{W}{4}$
	DN	Layer 1	$64 \times \frac{H}{4} \times \frac{W}{4}$
		Layer 2	$64 \times \frac{H}{4} \times \frac{W}{4}$
		Layer 3	$128 \times \frac{H}{8} \times \frac{W}{8}$
		Layer 4	$256 \times \frac{H}{16} \times \frac{W}{16}$
		GAP	$512 \times \frac{H}{32} \times \frac{W}{32}$
		FC Head	512
			1

from the strides in each layer. Factors like padding, kernel size and other parameters, while important, are not detailed in this overview for brevity and clarity.

In this scheme, the BN is split between Layer 0 and Layer 1; thus, forming SN constituted of Layer 0 and DN constituted of layers: {Layer 1, Layer 2, Layer 3, Layer 4, GAP, FC Head}. The DiffN is then attached between SN and DN forming thus the DiffConvNet. The DiffN is composed by a DiffLayer, which consists of a convolutional layer of 3×3 kernel size and an Averaging operation. The DiffLayer is repeated N times (the iteration number of diffusion), while Averaging operation averages the N diffused representations (initial, transient, and final states) into a unified diffused representation.

4. DeepFake case-study datasets

In order to guarantee diversity in our experimental simulations, we have used three different types of fake manipulation datasets namely, DeepFake Detection Challenge (DFDC), DeepFakeMnist+ (DFMNIST+) and EXPerts professional photoshop designers (EXP) datasets, and also a mixed one (MIXED), which is a mixed version of DFDC, DFMNIST+ and EXP.

DFDC dataset (Dolhansky et al., 2020) is currently the largest public facial forgery dataset being a collaborative endeavor by AWS, Facebook, Microsoft and the Partnership on AI's Media Integrity Steering Committee to counter the proliferation of deepfakes. Derived from GAN-generated fakes, it originally captured diverse individuals across various settings. For our experiments, we created a curated version¹ incorporating one image frame per video, isolating also the faces from every individual in the videos, comprising in total around 100000 instances.

DFMNIST+ (Huang et al., 2021) offers animated video data harnessed through the First Order Motion Model for Image Animation. It showcases 10,000 legitimate videos from VoxCeleb1, complemented by an equal number of animations stemming from the real videos. The animation videos vividly imitate a range of actions, from blinking to surprise and embarrassment, realized through a plethora of actors. Notably, to ensure quality, all animations successfully deceived contemporary liveness detectors, with prominent ones being from Baidu and Tianyan.

However, the "real" dataset did not have a unique actor ID for each instance; hence, in a 10-fold evaluation with random splits, the model risked on distinguishing based on persons instead of genuinely distinguishing between real and fake. This is because repeated appearances of the same individual across training and test sets could lead the model to recognize person-specific characteristics rather than discerning authenticity, introducing a potential bias. To sidestep this concern, we ensured each instance in our dataset represented a unique individual, preventing the model from resorting to person-specific recognition and compelling it to truly discern between real and fake.

The EXP dataset is a remarkable work of professional Photoshop designers of the Computational Intelligence and Photography Lab at Yonsei University's Computer Science Department. It uniquely features injected facial elements, such as eyes, noses, and mouths, transposed onto different faces. This dataset comprises around 1000 genuine and 1000 manipulated facial images, with the latter spanning varying degrees of detection difficulty. Leveraging EXP, we aspire to detect genuine fake-image markers, bypassing potential incidental patterns produced by GANs.

Aiming for a robust and comprehensive evaluation, we have also integrated these three distinct fake manipulation datasets. By combining these three datasets into a unified MIXED dataset, we aim to cover a wide range of manipulative techniques prevalent in the real world. This fusion not only adds diversity in the types of manipulations our model is exposed to but also enhances its generalization ability in different situations. Such a diverse training ensures that our model is not narrowly tailored to a specific kind of manipulation but is resilient against a variety of manipulation methods. Ultimately, this mixed-dataset approach reinforces our evaluation's reliability and underscores our model's adaptability in a more real-world scenario.

5. Comparison study

In this section, we present the experimental results from the comparison of the proposed model (DiffConvNet) with the state-of-the-art DeepFake detection approaches. Among them, a wide and diverse

range of different CNN-based methodologies were included (Fine-Tuning (Huang et al., 2021), Facial-Fusion (Tolosana et al., 2022), Facial-Geometry (Liang et al., 2023), Pairwise (Hsu et al., 2020), Siamese & Attention (Bonettini et al., 2021), Multi-View (Pintelas et al., 2023), Multi-Attentional (Zhao et al., 2021), and Distillation (Heo et al., 2021)), in order to perform comprehensible experimental comparisons.

5.1. Experimental setup

In this comparison the proposed approach was implemented with $N = 5$ (the effect of higher number of diffusion iterations were investigated in our ablation study in Section 6.2). In order to guarantee fairness in the experiments we have followed the following global experimental settings in all models described as follows:

- ResNet18 was selected as CNN backbone in all approaches because of its high balance between speed and accuracy (He et al., 2016; Canziani et al., 2017).
- The input dimensions of facial images for every dataset for every model was set to 256×256 . We must clarify that, the raw frames per video instance specifically for the DFDC dataset was much higher around to 1000×1000 . However since we cropped the faces per instance frame, the extracted facial images had much lower dimensions around to 256×256 .
- As regards the training parameters, Adam optimizer (Kingma and Ba, 2014) was used with an initial learning rate (10^{-4}) followed by reduce-learning-rate-on-plateau scheduler (Smith, 2017) in order to upgrade the learning rate with respect to the validation score (for every 5 epochs where there was no improvement, the learning was decreasing by a 0.7 factor). Additionally, early-stopping (Goodfellow et al., 2016) and weight-decay (Loshchilov and Hutter, 2019) were used in order to prevent over-fitting. In particular for early stopping, if there was no validation improvement for 10 epochs the training was terminated, while for weight decay, a coefficient of 10^{-4} was applied to the L2 regularization term, which penalizes large weights in the models.
- For solidness and robustness in the outcome results, 10-fold cross validation was also applied in all experimental evaluations. More specifically, 10 distinct folds were created comprised by a 90% training and 10% testing ratio each. Also, for each training fold, a separate validation set was also used (10% ratio of the remaining 90% training set fold) in order to apply the reduce-learning-rate-on-plateau scheduler and early stopping strategies described before. The final testing results were extracted by averaging the testing accuracy of each distinct testing fold (10 in total).
- The evaluation was based on the following performance metrics: Accuracy (Acc) in %, Area Under Curve (AUC), and Geometric Mean (GM) score. Acc is a straightforward method, which provides a direct measure of correct predictions but may be misleading for imbalanced datasets. AUC takes into account the probabilistic outputs of the model evaluating its ability to distinguish between classes across various thresholds offering higher evaluation depth beyond binary outcomes while GM is a robust metric, which measures the balance between sensitivity and specificity ensuring balanced performance across both classes (Pintelas and Pintelas, 2022).

It is worth mentioning that the numerical experiments and comparisons prioritize in the main concept and unique research value of each method. Our aim is to avoid adding extra complexities that are more suitable for real-world or competitions situations. Thus, we did not include advanced augmentations and backbones to boost performance, which might overshadow the true efficiency of our proposed method for deepfake detection.

¹ This version of this dataset can be found in the following [link](#)

Table 3

Performance comparison of DiffConvNet with state-of-the-art DeepFake detection methods across four datasets (DFDC, DFMNIST+, EXP, MIXED). Metrics include Accuracy (Acc), Area Under the Curve (AUC), and Geometric Mean (GM) with the best results highlighted in bold.

Approaches	DFDC			DFMNIST+			EXP			MIXED		
	Acc	AUC	GM	Acc	AUC	GM	Acc	AUC	GM	Acc	AUC	GM
Fine-Tuning	90.4	0.930	0.825	86.6	0.948	0.860	68.0	0.744	0.679	91.5	0.946	0.828
Facial-Fusion	93.2	0.959	0.847	87.5	0.946	0.867	70.9	0.783	0.707	91.9	0.953	0.832
Facial-Geometry	92.8	0.958	0.822	87.1	0.951	0.865	70.2	0.753	0.701	91.8	0.951	0.844
Pairwise	92.7	0.957	0.838	88.5	0.956	0.868	68.9	0.763	0.684	91.4	0.954	0.841
Siamese-Attention	91.4	0.951	0.837	87.1	0.940	0.862	67.1	0.754	0.670	91.3	0.953	0.855
Multi-View	87.8	0.934	0.845	85.2	0.928	0.838	71.1	0.778	0.709	91.6	0.955	0.840
Multi-Attentional	91.5	0.953	0.857	87.5	0.945	0.873	70.7	0.782	0.704	92.4	0.956	0.856
Distillation	92.8	0.956	0.832	88.2	0.949	0.875	69.9	0.758	0.697	92.6	0.954	0.843
DiffConvNet	92.9	0.962	0.860	88.5	0.953	0.877	71.2	0.791	0.711	91.7	0.959	0.864

Finally, in order to guarantee full transparency in the presented comparisons, the implementation code of the proposed model alongside with every other model can be found in.²

5.2. Numerical performance results

Table 3 summarizes the performance of all utilized DeepFake detection approaches applied on four facial manipulation datasets. Notice that the best performance for each metric is highlighted in bold.

As regards DFDC dataset, the proposed model exhibits the best overall performance, achieving an accuracy of 92.9%, AUC score of 0.962, and GM score of 0.860, indicating superior robustness in detecting DeepFakes on the DFDC dataset. Regarding the DFMNIST+ dataset, Pairwise approach reports the best performance relative to accuracy and AUC scores closely followed by DiffConvNet, which presented almost identical performance. In addition, the proposed model highlighted the best GM score (0.877), which implies that it exhibits the best accuracy balance across both classes.

In the context of the EXP dataset, the proposed model demonstrated superior performance compared to other state-of-the-art models, notably in the AUC score. Although the difference between the proposed model and the Multi-View is relative small in terms of accuracy (71.2% compared to 71.1%) and GM (0.711 compared to 0.709), our model achieved a substantial difference in terms of AUC (0.791 compared to 0.778). This performance indicates the model's efficiency in detecting professionally crafted manipulations, which are usually harder to detect, showcasing its utility in practical, real-world situations where high-quality manipulations are often encountered.

Finally, as regards the MIXED dataset, the proposed model demonstrates the best overall performance, reporting the highest score in two performance metrics, namely AUC (0.959) and GM (0.864). It is worth mentioning that the reported scores, provide strong empirical evidence about the considerable differences between the performance of the proposed model against the state-of-the-art models relative to both classes.

For providing strong statistical evidence about the effectiveness of the proposed DiffConvNet model, a statistical analysis is conducted for evaluating the hypothesis H_0 that all models present equally accurate predictions. For this purpose, non-parametric Friedman Aligned-Ranks (FAR) test (Hodges Jr. and Lehmann, 2011) is employed for ranking the models based on a performance metric; while Finner post-hoc test (Finner, 1993) with significance level $\alpha = 5\%$ is used for examining the existence of considerable differences and the presence of significant variations in their predictions, without any assumption about the distribution of the performance scores (Kiriakidou et al., 2024).

Tables 4–6 present the statistical analysis based on Acc, AUC and GM performance metrics, respectively. In all cases, the DiffConvNet reports the highest probability ranking based on FAR test. In addition,

Table 4

Statistical analysis: FAR and Li post-hoc tests results relative to Acc metric.

Model	Friedman	Finner post-hoc test	
	Ranking	p-value	H_0
DiffConvNet	8.125	–	–
Facial-Fusion	10.875	0.00596	Rejected
Distillation	12.875	0.000002	Rejected
Multi-Attentional	16.125	0.0	Rejected
Facial-Geometry	17.000	0.0	Rejected
Pairwise	18.625	0.0	Rejected
Multi-View	23.750	0.0	Rejected
Siamese-Attention	28.875	0.0	Rejected
Fine-Tuning	30.250	0.0	Rejected

Table 5

Statistical analysis: FAR and Li post-hoc tests results relative to AUC metric.

Model	Friedman	Finner post-hoc test	
	Ranking	p-value	H_0
DiffConvNet	6.250	–	–
Facial-Fusion	13.625	0.0	Rejected
Multi-Attentional	14.750	0.0	Rejected
Pairwise	15.625	0.0	Rejected
Distillation	18.875	0.0	Rejected
Facial-Geometry	19.750	0.0	Rejected
Multi-View	22.500	0.0	Rejected
Siamese-Attention	25.875	0.0	Rejected
Fine-Tuning	29.250	0.0	Rejected

Table 6

Statistical analysis: FAR and Li post-hoc tests results relative to GM metric.

Model	Friedman	Finner post-hoc test	
	Ranking	p-value	H_0
DiffConvNet	3.250	–	–
Multi-Attentional	8.375	0.0	Rejected
Facial-Fusion	16.875	0.0	Rejected
Distillation	19.250	0.0	Rejected
Multi-View	20.500	0.0	Rejected
Facial-Geometry	21.750	0.0	Rejected
Siamese-Attention	22.750	0.0	Rejected
Pairwise	23.000	0.0	Rejected
Fine-Tuning	30.750	0.0	Rejected

Li post-hoc test presents that there exist significant differences between the performance of DiffConvNet and the rest models, which highlights the effectiveness and prediction accuracy of the proposed model.

Summarizing, the presented numerical experiments highlight that the proposed model presented the best overall performance. In addition, they indicate the model's reliability and efficiency in distinguishing real from manipulated content effectively and its ability to handle a wide range of manipulations, reinforcing its robustness to generalize effectively across diverse types of facial manipulations.

² <https://github.com/EmmanuelPintelas/A-Diffusion-CNN-model-for-DeepFake-Image-Detection>

Table 7

Analysis of the effect of splitting point depths in the DiffConvNet model. Depth 0 corresponds to applying the diffusion network directly on the raw input, while Depth 1–4 refer to progressively deeper splits within the backbone.

DiffConvNet	Acc	AUC	GM
Depth 0	90.2	0.928	0.812
Depth 1	91.7	0.959	0.864
Depth 2	92.0	0.957	0.839
Depth 3	92.3	0.956	0.817
Depth 4	89.5	0.929	0.790

6. Ablation study

In this section, we engage a detailed exploration of the effectiveness of the proposed approach through an ablation study, which is structured into two categories: *Splitting-depth impact* and *Accuracy-speed trade-off*. Each of these studies investigates specific aspects of the proposed approach, providing insights into its effectiveness and adjustability. For the purpose of this analysis, we choose the MIXED dataset due to its rich diversity in manipulation techniques, since this dataset is a synthesis of the three major utilized dataset sources: DFDC, DFMNIST+, and EXP; making it an ideal choice for a broad and comprehensive assessment.

6.1. Splitting-depth impact

In this study, we examine the influence of different depths for the base network's (BN) splitting point. These depths range from the initial image (Depth 0) to deeper layers (Depth 1–4). Specifically, Depth 0 means that the DiffN is applied directly to the raw input image. Depth 1 specifies that the DiffN is integrated between Layer 0 and Layer 1, Depth 2 between Layer 1 and Layer 2, and so forth. For a more comprehensive understanding of the network's modules and layers, we refer to Table 2.

Based on the presented results in Table 7, it is evident that Depth 1 achieves the most balanced and superior results across the evaluated metrics, which balances between preserving the inherent coherence of the image and leveraging the feature insights provided by the early transformations of base network.

Consequently, Depth 1 is considered the most optimal depth for integrating the diffusion network and is adopted as the standard configuration for all experiments in this study. This depth demonstrates the efficacy of incorporating diffusion processes at initial layers, allowing for meaningful representations and feature extraction while retaining the essential characteristics and coherence of the input image.

6.2. Accuracy-speed trade-off

This study delves into the applicability of the proposed method across various CNN backbones. Although we primarily used ResNet for the experimental analysis because of its high balance between speed and accuracy, our methodology is not restricted to this backbone. Furthermore, we employed the proposed approach to other popular CNN architectures, namely Xception (Tolosana et al., 2022) and EfficientNet (EfficientNet-B0) (Raza et al., 2023), which are frequently applied in DeepFake detection. Finally, we conducted an Accuracy-Speed Trade-off analysis by integrating our diffusion-based approach into these utilized CNN backbones in order to examine the impact of various number of diffusion iterations N with respect to the prediction performance and the time overhead (total inference prediction time in seconds given a batch of instances).

Table 8 presents the performance results for the proposed DiffConvNet model, applied to various CNN backbones, with the diffusion iterations set to $N = 5$. Clearly, the incorporation of the diffusion

Table 8

Performance results of proposed DiffConvNet ($N = 5$) approach applied on various CNN backbones.

Models	Evaluation Metrics			CPU time (in secs)
	Acc	AUC	GM	
ResNet18	91.5	0.946	0.828	0.004
Diffusion - ResNet18	91.7	0.959	0.864	0.005
Xception41	90.8	0.945	0.843	0.014
Diffusion - Xception41	92.7	0.962	0.853	0.015
EfficientNet	92.9	0.964	0.858	0.013
Diffusion - EfficientNet	93.1	0.967	0.879	0.014

process enhances the performance across all utilized backbones, substantiating the efficacy of the proposed methodology in improving deepfake detection. The models exhibit improvement in all metrics especially in AUC and GM, showcasing additionally the approach's effectiveness.

As regards the computation cost, based on Table 8, there is approximately a 7% increase in time overhead for medium-heavy backbones like Xception and EfficientNet. This implies that the integration of the diffusion process does not significantly amplify the computational complexity of already complex and large networks, rendering the approach practical for more advanced models. However, for lighter models like ResNet18, there is a substantial, around 25% increase in time overhead. This underscores that while the methodology is generally beneficial, the impact on CPU time is more pronounced for lightweight models due to the relative increase in complexity.

In summary, these results indicate a well-balanced trade-off with the observed performance improvement leading to the fact that the increase in computational demand is worthy due to the significant performance gains.

Fig. 2 illustrates the trade-off between AUC performance and computation time of the three distinct utilized CNN backbones: ResNet18, Xception41, and EfficientNet, each enhanced with varying numbers of diffusion iterations N . Each point on the graph represents a distinct count of diffusion iterations ($N = 0, 5, 10, 20, 30, 40, 50$) incorporated into each backbone (the $N = 0$ refers to the initial standard CNN backbone without incorporating our proposed approach). The x -axis signifies the computation time, showcasing the overhead introduced by the diffusion process, while the y -axis depicts the AUC performance.

The interpretation of Fig. 2 suggests that both EfficientNet and ResNet demonstrate a superior balance between performance and computation time compared to Xception when incorporating the proposed diffusion methodology. As the number of diffusion iterations increases the computation time overhead for EfficientNet remains relatively stable achieving also the highest performance. On the other hand, ResNet, given its lightweight architecture, experiences a significant increase in computation time via the incorporation of diffusion process. However, it is noteworthy that the integration of the diffusion process into ResNet results in significant performance gains (enabling it to achieve the performance levels of EfficientNet), in contrast to EfficientNet which shows modest performance improvement across the increased diffusion iterations.

In summary, the findings of this ablation study imply to the fact that the proposed approach can be extended to a wide range of CNN backbones (compatibility property), improving also their prediction performance for identifying deepfakes (performance property), followed by a stable time overhead cost as N increases especially for larger and more complex architectures (scalability property).

7. Visualization study

Last but not least, we also conducted comprehensive visualization studies in order to enhance trust providing explanations and reasoning of the proposed model's efficacy that are meaningful and

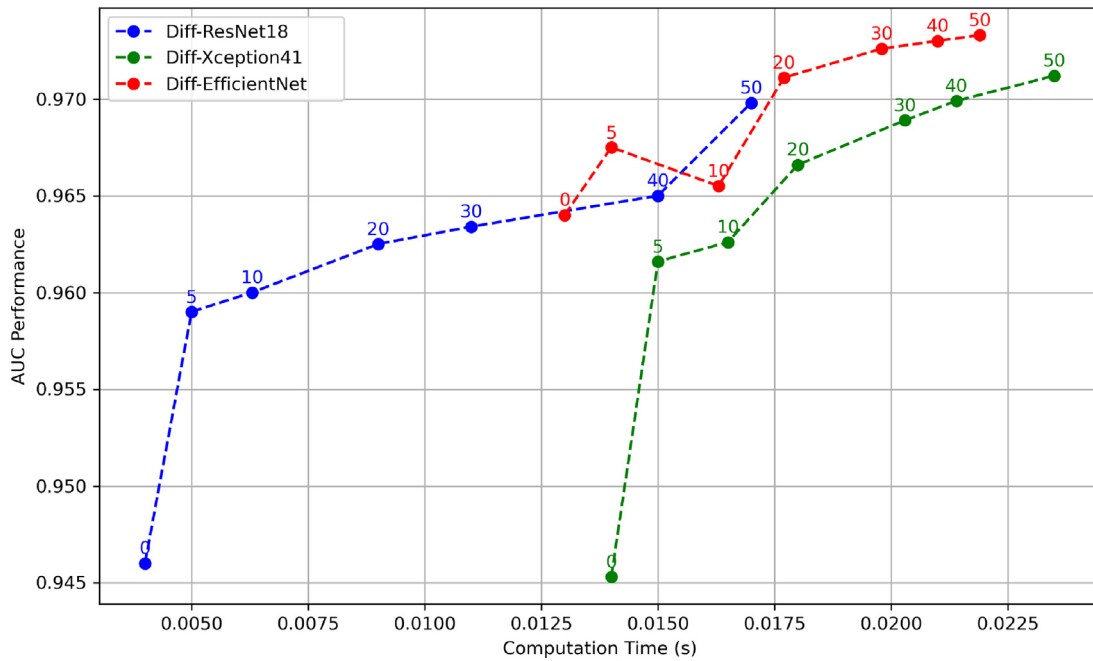


Fig. 2. Analysis of the accuracy-speed trade-off for different CNN backbones (ResNet18, Xception41, EfficientNet) integrated with the proposed diffusion methodology. Each point corresponds to a specific number of diffusion iterations (N), demonstrating the balance between computational cost and performance.

closer to human understanding, rather than relying solely on numerical outcomes. Specifically, we conducted interpretability evaluations and visualization of the diffusion states, for our proposed approach.

7.1. Predictions' interpretation

In order to enhance the reliability and trustfulness of our model, we performed a Grad-Cam-based (Livieris et al., 2023; Selvaraju et al., 2017) interpretation of predictions (Fig. 3), applied on the Exp dataset (due to the fact that the exact locations where the manipulations took place³ in the facial images are known). For this task, we presented various “extreme” case study instances for visual evaluation to further emphasize our point. These instances are challenging because what appears “real” might seem “fake” and vice versa.

The primary objective is to demonstrate that the model accurately focuses on the appropriate regions during its decision-making process. In contexts where specific manipulated regions are known (as with the Exp dataset), it is vital to ensure that the model’s attention correctly aligns with these areas to establish its reliability (Pintelas et al., 2021). It is essential thus to verify that the model concentrates on at least one manipulated region when predicting fake instances, which serves as a crucial indicator of its effectiveness and trustworthiness in detecting manipulations. If the model instead, focuses on irrelevant regions (e.g., the model focuses on hair, while the manipulation took place in eyes and mouth), its reliability would be questionable, even if the prediction is technically correct (Pintelas et al., 2020).

For image (C), depicted in Fig. 3, where manipulations are exclusively in the eyes (1100), the model notably focuses on these specific regions to classify the image as “fake”. Whereas in image (B) which features manipulations in both the eyes and nose (1110), the model’s attention focus more broadly, encompassing both eyes and the nose. In addition, in the case of image (D) (1111), the model’s focus extends beyond the eyes and nose, also encompassing the mouth area. This

observation is a positive indication given the known alterations across the eyes, nose, and mouth regions in this image.

Furthermore, for the real image (E), one can initially suspect manipulation due to the slightly asymmetrical and darker appearance of the left eye. So, the model properly focuses on this potential anomaly to determine whether it is a manipulation or not, ultimately making the correct decision that the image is real. Unfortunately, the model misinterpreted the image (F). The natural makeup on the subject’s face misled the model, primarily because the makeup around the left eye looks like to a possible fake manipulation, causing it to be misclassified as “fake”.

7.2. Diffusion states' interpretation

Finally, we also attempted to visualize the transient diffusion states from 0 to 50 of the diffusion network process (for $N = 50$ iterations), applied to the encoded representation of SN, using a ResNet18 backbone with a splitting point at Depth 1 (as described in Table 2). This allows us to illustrate the diffusion impact of our approach on some case study instances, as presented in Fig. 4.

We recall, that DeepFakes may show either unusually rapid diffusion states due to the artificial elements merging or exhibit strange diffusion patterns as they might not preserve the inherent coherence of a natural image. The primary objective of the diffusion approach is to discern potential distinctions between these scenarios. Through these visualizations, our aim is to provide empirical evidence by exploring whether discernible patterns emerge during the diffusion process, allowing for a more intuitive understanding of the proposed method beyond solely relying on numerical experimental results. However, it is important to acknowledge that such patterns may be subtle, ambiguous, or even imperceptible to the human eye. Our goal is not to claim that these visuals definitively expose DeepFakes but rather to demonstrate how the diffusion process refines and enriches the initial encoded representation, potentially aiding a CNN model in detecting DeepFakes more effectively.

As presented in Fig. 4, for the “real” cases, the transition through each transient diffusion state appears smoother, more natural and more stable. In contrast, the “fake” cases exhibit more abrupt and unnatural

³ Each manipulation is denoted by a binary four-digit code indicating which parts of the face were altered (left eye, right eye, nose, mouth). For instance, the code ‘1100’ signifies that both eyes were manipulated, while the nose and mouth remain original.

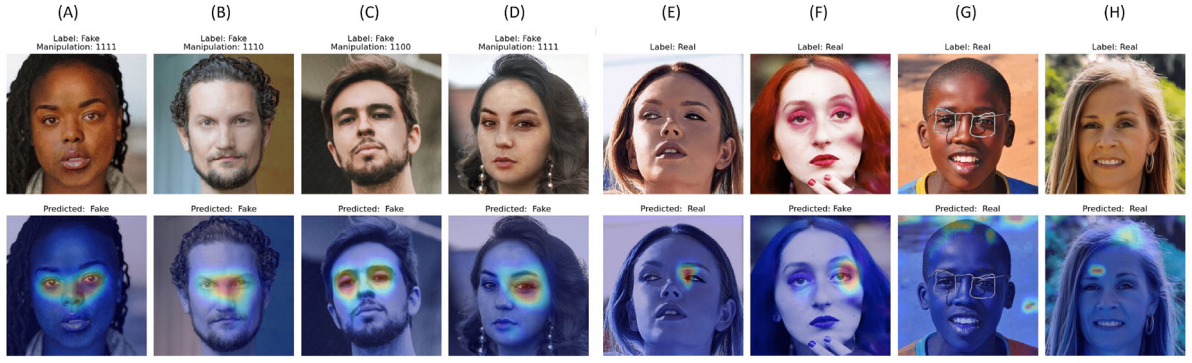


Fig. 3. Grad-Cam visualization of the DiffConvNet model's predictions on real and manipulated facial images from the EXP dataset. The highlighted regions indicate the model's focus areas for decision-making, demonstrating its ability to detect manipulated regions effectively.



Fig. 4. Sequential visualization of the transient diffusion states for “real” and “fake” images across 50 iterations of proposed diffusion network. The diffusion process exposes distinct patterns, with manipulated images displaying abrupt transitions and inconsistencies compared to smoother transitions in authentic images.

transitions; for example in the transition from states 4 to 5 to 6 for the “fake” case (D).

Furthermore, the “fake” cases exhibit more odd and inconsistent diffusion patterns. For example, in the “fake” case (B), noticeable inconsistencies emerge in states 6, 8, 12, and 26, where unnatural structural deviations can be observed. In particular, for this case, the left eye appears inconsistent with the right eye, which appears significantly lighter/highlighted. This asymmetry strongly suggests a manipulation artifact/pattern, especially since we already know that the eyes of this person were manipulated by blending them with another person’s facial features³. In contrast to the “real” cases, where facial features are diffused more smoothly, preserving their overall spatial structure and coherence without too many deviations.

Additionally, we conducted two additional experiments to visually evaluate and compare alternative diffusion-like processes: (1) iterative Gaussian blurring and (2) Gaussian noise-based diffusion (Xiao et al., 2024). Both methods were also applied to the output of SN (using the same splitting point at Depth 1, refer Table 2), ensuring a fair baseline for comparison.

The comparison between the network-based diffusion process (Fig. 4) and the two Gaussian-based transformations (Figs. 5 and 6) highlight the distinct advantages of the proposed approach:

- Gaussian blurring (Fig. 5): This process progressively smooths the image, causing an indiscriminate loss of fine-grained details. Both real and fake images undergo similar degradation, preventing it from exposing any deepfake-specific inconsistencies.
- Gaussian diffusion (Fig. 6): This method introduces progressive noise, leading to stochastic image degradation. Interestingly, some manipulation artifacts persist throughout the diffusion process (e.g., the left manipulated eye in the fake case (B)), which aligns with our hypothesis that manipulated images might exhibit distinct diffusion patterns due to their unnatural statistical properties. However, unlike the proposed diffusion network (Fig. 4), Gaussian diffusion lacks a learnable filtering mechanism. As a result, the process does not selectively highlight manipulated artifacts and inconsistencies to further reveal discriminative features.

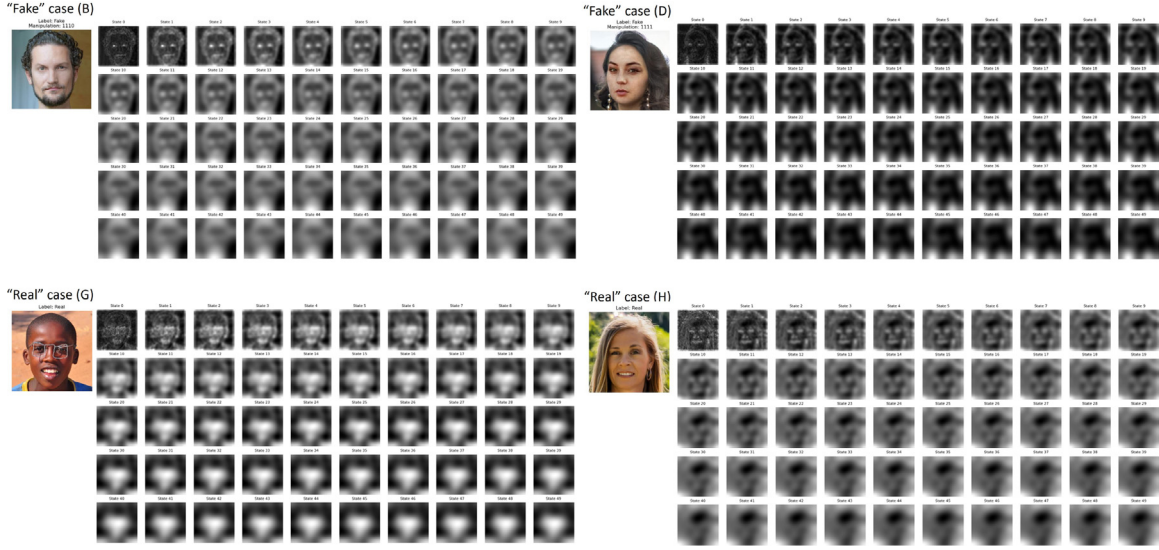


Fig. 5. Sequential visualization of iterative Gaussian blurring applied to “real” and “fake” images across 50 iterations. This serves as a baseline comparison to assess the proposed diffusion process’s effects. Gaussian blurring leads to a homogeneous loss of detail, lacking the discriminative features observed in the transitions of Fig. 4.

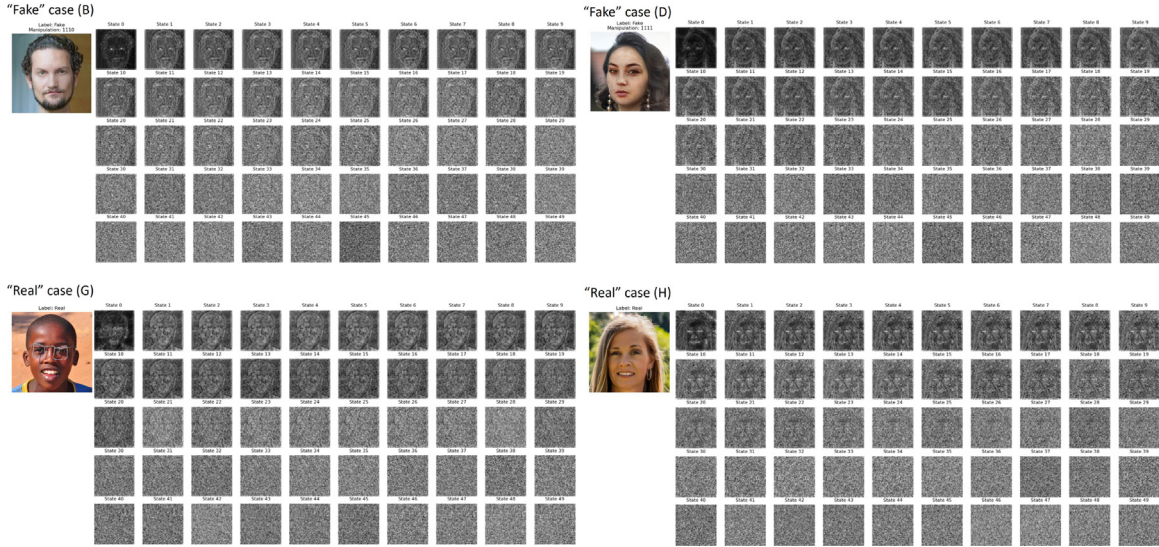


Fig. 6. Sequential visualization of Gaussian noise-based diffusion applied to “real” and “fake” images across 50 iterations. This method introduces progressive noise, revealing some manipulation artifacts due to unnatural statistical properties. However, unlike the proposed network-based diffusion approach, it lacks its learnable mechanism, preventing it from selectively highlighting deepfake inconsistencies and enhancing discriminative features.

These findings demonstrate that the proposed network-based diffusion framework is not merely a random noise operation, but a targeted diffusion mechanism acting as a learnable feature extraction and filtering process for DeepFake detection.

8. Discussion & reasoning of methodology

In this Section, we provide a thorough reasoning and discussion of the key procedures and methods used in the proposed approach. This includes the diffusion procedure and methodology in general, the rationale behind the initial fine-tuning procedure (step 1)⁴, the cutting/splitting-based approach (step 2), the decision on the splitting “depth” which involves whether to split shallowly or more deeply (step 4), the averaging of the diffused representation (step 6), and the final end-to-end DiffConvNet fine-tuning (step 8).

⁴ Steps 1–8 refer to the steps of Fig. 1 of Section 3.

As regards the diffusion procedure, our approach is motivated by the intuition that deepfakes could exhibit unusual diffusion states due to unnatural artificial blending. However, we consider essential to clarify that the diffusion process could not perform just by itself all the “hard job” for exposing the deepfake manipulations; it also requires an efficient image encoding for performing diffusion and an efficient exploitation of the diffusion process, in order to extract patterns and features. In fact, the diffusion process mainly aims on assisting a CNN classifier (DN) to detect deepfakes by providing it with a “richer” representation (diffused-representation: d), compared to providing only the encoded feature maps (x_0) of the initial image as extracted via the shallow convolutional layers of a standard CNN backbone.

Regarding the rationale of the method used for creating diffusion process, it is described as follows: Diffusion processes, fundamental to many physical and mathematical models, are typically represented using Partial Differential Equations (PDEs) (Gilboa and Osher, 2009; Osher and Shen, 2019). Traditional methods for simulating these processes have often employed techniques derived from

PDEs, such as Gaussian diffusion (Xiao et al., 2024), Laplacian diffusion (Osher and Shen, 2019), anisotropic diffusion (Tsotsios and Petrou, 2013), and bilateral filtering (Paris et al., 2009). Instead, we adopted a convolutional-based approach allowing for a flexible, efficient, and highly parallelizable diffusion simulation. Convolutional layers can be interpreted as discrete approximations to PDEs, since in a convolution operation, the filters (or kernels) serve functions analogous to the spatial derivatives in PDEs (Ruthotto and Haber, 2020; Kag and Saligrama, 2022). Therefore, interpreting convolutional operations as approximations to PDEs it is a logical step to suggest that convolutional layers can simulate diffusion processes.

Regarding the rationale of the initial fine-tuning procedure, it is described as follows: Since we cut/split the BN into two sub-networks (SN and DN), the initial fine-tuning procedure of BN aims to ensure that the SN will encode the input image into a more relevant encoded representation x_0 with respect to the deepfake task. Thus, the diffusion input x_0 may have already highlighted some initial deepfake patterns, while now since the diffusion process will start its operation from a better initial point, it can perform more “targeted” diffusion on x_0 , for augmenting even more the deepfake’s feature extraction procedure; hence, providing more enriched data representations for the DN classifier.

As regards the reasoning for adopting the splitting-based approach, we clarify it as follows: Instead of using separate custom-designed Shallow Network (SN) and Deep Network (DN), we aim to leverage the prior knowledge encapsulated in a pre-trained CNN backbone, such as ResNet. Consequently, we strategically split this backbone into two sub-networks: the SN, tasked with encoding input images into compact, robust, and noise-reduced latent representations ideal for targeted diffusion, and the DN, specialized in extracting high-level features from the diffusion outcomes and performing the final classification. This methodology provides significant advantages in terms of functionality, adjustability, and performance since our diffusion-based strategy can seamlessly integrate with various CNN backbones, fully exploiting their pre-trained weights.

However, the question of identifying an optimal splitting depth arises naturally: Why not directly apply diffusion to raw input images, or why not utilize deeper latent representations? The rationale behind our selected splitting point involves carefully balancing two critical factors: retaining the inherent natural coherence of images and effectively extracting meaningful representations for diffusion. Splitting the network at excessively deep layers would risk losing essential coherent structural information inherent to natural images, rendering the diffusion process inefficient. Our method fundamentally relies on detecting unnatural diffusion patterns characteristic of DeepFakes, which requires preserving initial image coherence. On the other hand, applying diffusion directly at the raw pixel level (depth 0) would be significantly more computationally expensive and less efficient, as raw images contain redundant information and irrelevant noise, bypassing valuable preliminary structural transformations performed by the initial network layers. To empirically justify this choice, our ablation study (Section 6.1) presents a comprehensive performance analysis across multiple splitting depths, ranging from raw input images (depth 0) to progressively deeper encoded layers (e.g., depths 1 and 2).

For the decision to average the diffused representations, the primary motivations are for ensuring robustness and preserving the structure and weight configuration of the used CNN backbone. More specifically, by averaging the transient diffusion states, the model minimizes the effect of noise and potential outliers that might have been introduced during the diffusion process. This approach effectively works as an ensemble method within the diffusion layers, aggregating the outcomes and thereby increasing stability and robustness.

Also, the diffusion process, by its very nature, captures varying feature scales and patterns across different states. Averaging them aims to fuse these diverse feature representations into a unified feature map. This operation ensures that the network benefits from the rich patterns

captured across different diffusion states, resulting in a more comprehensive and integrated feature representation for the subsequent layers of the DN.

Furthermore, it is necessary to ensure that the input representation for the DN matches the ancestral BN architecture. This is particularly crucial as regards the adaptability property to maintain the weight structure and more importantly, the integrity of the utilized CNN backbone. By averaging the diffusion states, the network maintains a consistent input dimensionality that aligns with the requirements of the downstream modules in the DN. For example, in our context, the DiffN, given an encoded representation x_0 of dimensions $64 \times \frac{H}{4} \times \frac{W}{4}$, creates a $N \times 64 \times \frac{H}{4} \times \frac{W}{4}$ representation (Table 2), which does not match the $64 \times \frac{H}{4} \times \frac{W}{4}$ input dimensions of DN. Thus, by averaging the diffused representation we ensure that the output of DiffN will match the input dimensions of DN preserving thus the CNN backbone’s architecture.

Finally, as regards the final end-to-end DiffConvNet model’s fine-tuning with respect to the DeepFake classification loss, one basic reason is to optimize the DN to adjust to its new input representation. Originally, the DN was fine-tuned via the BN’s initial fine-tuning process, adapted to identify patterns from the x_0 representation. However, with its new input being the diffused representation d , it will have to re-adjust its weights to discern the patterns lied into this new representation. Also, we let the SN and DiffN weights to get updated. This aims to let the SN to learn an even better encoded representation for diffusion further optimized via the end-to-end training, while since for the diffusion process we do not use a fixed diffusion function like gaussian, we let instead the model, via the end-to-end training, to find its own diffusion function which will be more optimal regarding the deepfake detection task.

9. Conclusions

This work focuses on image-based DeepFake detection. Although deepfakes are predominantly recognized as manipulated videos, there exist many cases to be produced as manipulated images (Nightingale et al., 2017; Yan et al., 2018). Furthermore, the world has witnessed cases where deepfakes are mainly manifested in image form, especially in scenarios where a brief but impactful disinformation is intended, without the need for longer video narratives (Chesney and Citron, 2019). Such instances underscore the importance of an effective image-based deepfake detection method. Additionally, the challenge of detecting deepfakes in images is arguably more complex because the manipulations or “injections” are harder to be detected accurately in a static frame as opposed to a dynamic video sequence. As regards video-based detection, the manipulations can be more easily exposed due to the plethora of frames and the continuity of motion (Thies et al., 2016).

In addition, it is essential to mention that although this work focuses on image-based deepfake detection, the proposed framework could be easily extended into deepfake videos. This is because the transition from image to video essentially involves processing sequences of images (frames) and thus our image-based model could be easily applied to individual frames of a video, and then the model’s outputs for each frame could be aggregated to derive a final prediction. Therefore, if a model achieves satisfactory performance in image-based deepfake detection, it is reasonable to infer that it would do so in video-based detection too.

In this study, we proposed a CNN-based model, DiffConvNet, which leverages diffusion processes to detect DeepFake manipulations in images. The approach is based on the hypothesis that manipulated images exhibit distinct diffusion patterns due to their unnatural distributions introduced by generative processes. The diffusion mechanism aims to expose inconsistencies within the input image, enhancing the CNN’s ability to identify DeepFake content. Extensive experimental evaluations demonstrated the effectiveness of this framework, highlighting its potential as a promising tool in the field of DeepFake detection. In conclusion, the main advantages of the proposed approach are summarized as follows:

- **Accuracy:** The model demonstrates high performance across a wide range of facial image manipulations (refer to Table 3).
- **Compatibility & Performance:** The approach exhibits adaptability with various CNN backbones that also manage to significantly improve their performance (see Sections 6.2 and 8).
- **Scalability:** Our framework can be efficiently scaled to larger and more complex architectures (refer to Section 6.2).
- **Interpretability:** Incorporating diffusion patterns can provide better insight into the detection process, allowing for more transparent and trustworthy DeepFake decision (refer to Section 7).

However, the main disadvantage of our approach lies in the increased computation time compared to standard CNN backbones (Table 8), particularly when the model has to handle a large number of diffusion iterations, such as $N = 50$ (Fig. 2). This is a trade-off that our approach needs to deal with in order to create an efficient diffusion simulation for exposing manipulation content, especially for the very high-quality manipulated image cases, such as those depicted in Figs. 3(B) and 3(C), which can be considered very hard to detect.

Nevertheless, as with any methodology, our study has limitations that warrant further investigation. The full potential of transient diffusion states remains unexplored. While our averaging approach aimed to create a robust diffused representation compatible with various CNN backbones, more advanced strategies should be considered. For instance, integrating CNN-LSTM-based methods (Köpüklü et al., 2022; Jain et al., 2020) could capture sequential and dynamic feature correlations across transient diffusion states. Additionally, exploring alternative diffusion architectures and sampling strategies (Yang et al., 2023), may enhance both performance and efficiency.

CRedit authorship contribution statement

Emmanuel Pintelas: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ioannis E. Livieris:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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